



Networking in Linux

Connecting to the interwebs

Networking forms the backbone of modern computing, enabling seamless communication between systems. Whether configuring a simple home connection or managing an enterprise-grade network, Linux offers a vast array of tools and utilities to facilitate robust and secure networking. With its deeply integrated network stack, Linux provides administrators with unparalleled control over network configuration, remote access, security, and diagnostics.

This chapter explores the fundamentals of networking in Linux, diving into configuration methods, security mechanisms, and diagnostic tools. By the end, you will have a firm grasp of how to configure and troubleshoot Linux networks effectively. Understanding these aspects is crucial for system administrators, developers, and even everyday users who rely on Linux for various networking tasks.

Network Configuration

IP Addressing: Static vs. DHCP (nmcli, netplan)

Every device connected to a network requires an IP address, which can either be assigned statically or dynamically. While static addressing offers control and predictability, DHCP simplifies management, especially in larger networks where manually assigning addresses would be cumbersome.

Static IP addressing is often used for servers, printers, and other critical devices that require a fixed network identity. DHCP, on the other hand, is managed by a DHCP server, dynamically assigning IP addresses to devices as they connect to the network.

Configuring Static IP with nmcli

nmcli (NetworkManager Command Line Interface) allows